

Computer Networks (CN)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 35

Question 1 of 35 | GATE 2025 Set 1 (CS1) | Q.16 (Computer Science & IT)

Q.16

Identify the ONE CORRECT matching between the OSI layers and their corresponding functionalities as shown.

OSI Layers

Functionalities

(a) Network layer

(I) Packet routing

(b) Transport layer

(II) Framing and error handling

(c) Datalink layer

(III) Host to host communication

(A) (a)-(I), (b)-(II), (c)-(III)

(B) (a)-(I), (b)-(III), (c)-(II)

(C) (a)-(II), (b)-(I), (c)-(III)

(D) (a)-(III), (b)-(II), (c)-(I)

Question 2 of 35 | GATE 2025 Set 1 (CS1) | Q.22 (Computer Science & IT)

Q.22

Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

(A) P3: SYN = 1, ACK = 1

(B) P2: SYN = 1, ACK = 1

(C) P2: SYN = 0, ACK = 1

(D) P1: SYN = 1

Question 3 of 35 | GATE 2025 Set 1 (CS1) | Q.26 (Computer Science & IT)

Q.26

Which of the following statement(s) is/are TRUE for any binary search tree (BST) having n distinct integers?

(A) The maximum length of a path from the root node to any other node is $(n-1)$.

- (B) An inorder traversal will always produce a sorted sequence of elements.
- (C) Finding an element takes $O(\log_2 n)$ time in the worst case.
- (D) Every BST is also a Min-Heap.

Question 4 of 35 | GATE 2025 Set 1 (CS1) | Q.40 (Computer Science & IT)

Q.40

A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

- (A) E3
- (B) E1
- (C) E2
- (D) E5

Question 5 of 35 | GATE 2025 Set 2 (CS2) | Q.16 (Computer Science & IT)

Q.16

Consider the following statements:

(i)

Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address.

(ii)

A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S.

Which ONE of the following is CORRECT?

- (A) Both (i) and (ii) are TRUE
- (B) (i) is TRUE and (ii) is FALSE
- (C) (i) is FALSE and (ii) is TRUE
- (D) Both (i) and (ii) are FALSE

Question 6 of 35 | GATE 2025 Set 2 (CS2) | Q.17 (Computer Science & IT)

Q.17

Consider the routing protocols given in List I and the names given in List II:

List I

List II

(i)

Distance vector routing

(a)

Bellman-Ford

(ii)

Link state routing

(b)

Dijkstra

For matching of items in List I with those in List II, which ONE of the following options is CORRECT?

(A) (i) – (a) and (ii) – (b)

(B) (i) – (a) and (ii) – (a)

(C) (i) – (b) and (ii) – (a)

(D) (i) – (b) and (ii) – (b)

Question 7 of 35 | GATE 2025 Set 2 (CS2) | Q.23 (Computer Science & IT)

Q.23

Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes.

Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?

(A) 5 fragments, 1488 bytes

(B) 6 fragments, 88 bytes

(C) 6 fragments, 108 bytes

(D) 6 fragments, 116 bytes

Question 8 of 35 | GATE 2024 Set 1 (CS1 S5) | Q.29 (Computer Science & IT)

Q.29

TCP client P successfully establishes a connection to TCP server Q. Let NP denote the sequence number in the SYN sent from P to Q. Let NQ denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

(A) The sequence number NP is chosen randomly by P

(B) The sequence number NP is always 0 for a new connection

(C) The acknowledgement number NQ is equal to NP

(D) The acknowledgement number NQ is equal to NP+ 1

Question 9 of 35 | GATE 2024 Set 1 (CS1 S5) | Q.31 (Computer Science & IT)

Q.31

Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

- (A) Source IP
- (B) Destination IP
- (C) Header Checksum
- (D) Total Length

Question 10 of 35 | GATE 2024 Set 1 (CS1 S5) | Q.58 (Computer Science & IT)

Q.58

Consider the entries shown below in the forwarding table of an IP router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation “/N” in a prefix indicates a subnet mask with the most significant N bits set to 1.

Prefix

Next hop router

10.1.1.0/24

R1

10.1.1.128/25

R2

10.1.1.64/26

R3

10.1.1.192/26

R4

This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are 10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191, and 10.1.1.205. The number of packets forwarded via the next hop router R2 is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 11 of 35 | GATE 2024 Set 1 (CS1 S5) | Q.60 (Computer Science & IT)

Q.60

The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 12 of 35 | GATE 2024 Set 2 (CS2 S6) | Q.23 (Computer Science & IT)

Q.23

Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

- (A) The destination IP address is the IP address of R
- (B) The destination IP address is the IP address of Y
- (C) The destination MAC address is the MAC address of S
- (D) The destination MAC address is the MAC address of Y

Question 13 of 35 | GATE 2024 Set 2 (CS2 S6) | Q.32 (Computer Science & IT)

Q.32

Which of the following fields of an IP header is/are always modified by any router before it forwards the IP packet?

- (A) Source IP Address
- (B) Protocol
- (C) Time to Live (TTL)
- (D) Header Checksum

Question 14 of 35 | GATE 2024 Set 2 (CS2 S6) | Q.54 (Computer Science & IT)

Q.54

Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 15 of 35 | GATE 2024 Set 2 (CS2 S6) | Q.55 (Computer Science & IT)

Q.55

Consider an Ethernet segment with a transmission speed of 108 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (in bits) required for collision detection is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 16 of 35 | GATE 2023 Single Session | Q.24 (Computer Science & IT)

Q.24

Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?

- (A) OSPF implements Bellman-Ford algorithm to find shortest paths.
- (B) OSPF uses Dijkstra's shortest path algorithm to implement least-cost path routing.
- (C) OSPF is used as an inter-domain routing protocol.
- (D) OSPF implements hierarchical routing. CS

Question 17 of 35 | GATE 2023 Single Session | Q.64 (Computer Science & IT)

Q.64

The forwarding table of a router is shown below.

Subnet Number

Subnet Mask

Interface ID

200.150.0.0

255.255.0.0

1

200.150.64.0

255.255.224.0

2

200.150.68.0

255.255.255.0

3

200.150.68.64

255.255.255.224

4

Default

0

A packet addressed to a destination address 200.150.68.118 arrives at the router.

It will be forwarded to the interface with ID

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 18 of 35 | GATE 2022 Single Session | Q.22 (Computer Science & IT)

Q.22

Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below.

What is the number of subnets inside the enterprise network?

- (A) 3
- (B) 12
- (C) 6
- (D) 8

Question 19 of 35 | GATE 2022 Single Session | Q.55 (Computer Science & IT)

Q.55

Consider routing table of an organization's router shown below:

Subnet Number

Subnet Mask

Next Hop

12.20.164.0

255.255.252.0

R1

12.20.170.0

255.255.254.0

R2

12.20.168.0

255.255.254.0

Interface 0

12.20.166.0

255.255.254.0

Interface 1

default

R3

Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

- (A) 12.20.164.0/20
- (B) 12.20.164.0/22
- (C) 12.20.164.0/21
- (D) 12.20.168.0/22

Question 20 of 35 | GATE 2022 Single Session | Q.59 (Computer Science & IT)

Q.59

Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3×10^8 m/s. The time taken (in milliseconds, rounded off to two decimal places) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 21 of 35 | GATE 2022 Single Session | Q.60 (Computer Science & IT)

Q.60

Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 22 of 35 | GATE 2019 Single Session | Q.16 (Computer Science & IT)

Q.16 Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

- (A) IMAP, POP3
- (B) SMTP, POP3
- (C) SMTP, MIME
- (D) IMAP, SMTP CS 4/15

Question 23 of 35 | GATE 2019 Single Session | Q.28 (Computer Science & IT)

Q.28 Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true?

- (A) M, N, and P all belong to the same subnet
- (B) Only M and N belong to the same subnet
- (C) Only N and P belong to the same subnet
- (D) M, N, and P belong to three different subnets

Question 24 of 35 | GATE 2019 Single Session | Q.49 (Computer Science & IT)

Q.49 Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 25 of 35 | GATE 2018 Single Session | Q.13 (Computer Science & IT)

Q.13 Match the following:

Field

Length in bits

P. UDP Header's Port Number

I. 48

Q. Ethernet MAC Address

II. 8

R. IPv6 Next Header

III. 32

S. TCP Header's Sequence Number

IV. 16

- (A) P-III, Q-IV, R-II, S-I
- (B) P-II, Q-I, R-IV, S-III
- (C) P-IV, Q-I, R-II, S-III
- (D) P-IV, Q-I, R-III, S-II CS 5/20

Question 26 of 35 | GATE 2018 Single Session | Q.14 (Computer Science & IT)

Q.14 Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that *cwnd* stands for the TCP congestion window and *MSS* denotes the Maximum Segment Size.

- (i) The *cwnd* increases by 2 *MSS* on every successful acknowledgment.
- (ii) The *cwnd* approximately doubles on every successful acknowledgement.
- (iii) The *cwnd* increases by 1 *MSS* every round trip time.
- (iv) The *cwnd* approximately doubles every round trip time.

Which one of the following is correct?

- (A) Only (ii) and (iii) are true
- (B) Only (i) and (iii) are true
- (C) Only (iv) is true
- (D) Only (i) and (iv) are true

Question 27 of 35 | GATE 2018 Single Session | Q.25 (Computer Science & IT)

Q.25 Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps (= 10⁹ bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 28 of 35 | GATE 2016 Set 1 (CS1) | Q.24 (Computer Science & IT)

Q.24

Which one of the following protocols is NOT used to resolve one form of address to another one?

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) RARP

Question 29 of 35 | GATE 2016 Set 1 (CS1) | Q.25 (Computer Science & IT)

Q.25

Which of the following is/are example(s) of stateful application layer protocols?

- (i) HTTP
- (ii) FTP
- (iii) TCP
- (iv) POP3

- (A) (i) and (ii) only
- (B) (ii) and (iii) only
- (C) (ii) and (iv) only
- (D) (iv) only CS(Set A) 8/17 GATE 2016

Question 30 of 35 | GATE 2016 Set 1 (CS1) | Q.52 (Computer Science & IT)

Q.52

Consider that B wants to send a message m that is digitally signed to A. Let the pair of private and public keys for A and B be denoted by K^- and K^+

x for $x = A, B$, respectively. Let $K_x(m)$

represent the operation of encrypting m with a key K_x and $H(m)$ represent the message digest.

Which one of the following indicates the CORRECT way of sending the message m along with the digital signature to A?

- (A) $\{m, K^+ B (H(m))\}$
- (B) $\{m, K^- B (H(m))\}$
- (C) $\{m, K^- A (H(m))\}$
- (D) $\{m, K^+ A (m)\}$

Question 31 of 35 | GATE 2016 Set 1 (CS1) | Q.53 (Computer Science & IT)

Q.53

An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes.

The number of fragments that the IP datagram will be divided into for transmission is

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 32 of 35 | GATE 2016 Set 1 (CS1) | Q.54 (Computer Science & IT)

Q.54

For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is seconds.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 33 of 35 | GATE 2016 Set 2 (CS2) | Q.25 (Computer Science & IT)

Q.25

Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

- (A) HTTP GET request, DNS query, TCP SYN
- (B) DNS query, HTTP GET request, TCP SYN
- (C) DNS query, TCP SYN, HTTP GET request
- (D) TCP SYN, DNS query, HTTP GET request

Question 34 of 35 | GATE 2016 Set 2 (CS2) | Q.36 (Computer Science & IT)

Q.36

Consider the following New-order strategy for traversing a binary tree:

- Visit the root;
- Visit the right subtree using New-order;
- Visit the left subtree using New-order;

The New-order traversal of the expression tree corresponding to the reverse polish expression

$3\ 4\ *\ 5\ -\ 2\ ^\ 6\ 7\ *\ 1\ +\ -$

is given by:

- (A) $+ - 1\ 6\ 7\ *\ 2\ ^\ 5 - 3\ 4\ *$
- (B) $- + 1\ * 6\ 7\ ^\ 2 - 5\ * 3\ 4$
- (C) $- + 1\ * 7\ 6\ ^\ 2 - 5\ * 4\ 3$
- (D) $1\ 7\ 6\ *\ + 2\ 5\ 4\ 3\ *\ -\ ^\ -$

Question 35 of 35 | GATE 2016 Set 2 (CS2) | Q.53 (Computer Science & IT)

Q.53

A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is bytes.

CS(Set B)

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]